[A11] Household Sustainability Challenge Platform - Project PROPOSAL

Group ID	Team Member Names	Student ID
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Executive Summary

This project will deliver a web-based Minimum Viable Product (MVP) for Sustainable Communities SA (SCSA) that helps households translate sustainability advice into measurable daily actions. Core deliverables include secure user accounts and privacy choices, goal and challenge management, manual activity and consumption logging, an interactive progress dashboard, transparent rule-based recommendations, achievement badges, aggregated community reporting, source code, testing, and handover documentation.The outcome will be an extensible, privacy-conscious prototype that SCSA can demonstrate, evaluate and continue developing.

1. Project Background and Description

Sustainable Communities SA (SCSA) is a non-profit organisation that promotes environmentally responsible living across South Australia. Although community interest in resource conservation is growing, households often struggle to sustain long-term habits because they lack practical tracking tools, measurable targets and timely feedback.

To address this gap, the project will develop the Household Sustainability Challenge Platform, an interactive web application for personal progress tracking and sustainability challenges. It will allow households to monitor energy, water and waste-related actions while retaining clear control over how their information is used. SCSA will gain aggregated, privacy-conscious insights that can support community engagement and demonstrate collective progress to stakeholders.

The project objectives are to help households set and monitor sustainability goals, provide clear evidence of progress, protect user choice over data sharing, and give SCSA aggregated insights that support programme evaluation without exposing personal records. Expected benefits include stronger participant engagement, more consistent habit tracking, clearer feedback and a maintainable foundation for future development.

2. Project Scope

The project scope is structured to deliver a functional Minimum Viable Product (MVP) prototype within the timeframe. The work focuses on core functionality, essential user workflows and high-impact visual analytics.

Inclusions

The following aspects of the project are considered within scope:

- **Requirements Analysis & UX/UI Prototyping:** Documenting user needs and acceptance criteria, mapping user workflows, and designing responsive interfaces for desktop and mobile web browsers.
- **User Account & Household Profile Management:** Implementing secure user registration, authentication and household profile settings. Records will be private by default, with separate opt-in choices for anonymous aggregated community reporting and optional pseudonymous leaderboard participation.
- **Goal & Challenge Engine:** Developing functional modules for users to select predefined sustainability targets for electricity, water and waste, create suitable personal goals, and join recurring green challenges such as a seven-day plastic-reduction challenge.
- **Manual Data Logging System:** Creating intuitive web forms for users to manually record quantitative utility metrics (kWh, litres and kg) and qualitative eco-actions such as public transport use.
- **Data Visualisation & Analytics Dashboard:** Integrating dynamic charts to display individual consumption trends, target progress and estimated emissions-equivalent savings calculated using documented factors.
- **Gamification & Community Reporting:** Implementing achievement badges and aggregated community reporting as baseline features. An optional pseudonymous leaderboard is a stretch feature and is not part of baseline acceptance.
- **Database Architecture & Back-end API:** Designing a relational database schema and application programming interface to store household records, activity histories and challenge configurations securely.

Exclusions

The following aspects of the project are explicitly out of scope:

- **Automated IoT or Smart Meter Integration:** Direct API or hardware integration with external utility providers (e.g., smart electricity/water meters) will not be built; all data entry will rely on manual user input.
- **Native Mobile Application Development:** The deliverable is a responsive web application. Dedicated native iOS or Android applications and cross-platform mobile builds are outside the project scope.
- **AI-Driven Automated Recommendation Engine:** Machine-learning and generative-AI recommendation engines are excluded. Any recommendation feature will use transparent, preconfigured rules tied to selected goals and recorded activities.
- **Third-Party Social Media API Integration:** Direct automated cross-posting to external platforms (e.g., Facebook, Instagram) is excluded.
- **Native In-App Messaging/Chat System:** Direct household-to-household real-time messaging capabilities will not be developed.

- **Production Server Hosting & Domain Deployment:** Commercial production hosting, operational support and long-term domain deployment are outside the project scope. Local or staging deployment and an optional temporary demonstration domain are permitted.
- **Multi-Language & Localization (i18n):** The platform interface and challenge content will be provided exclusively in English.
- **Multi-Factor Authentication (MFA) & Enterprise Identity Providers:** The baseline prototype will use standard email and password authentication. Single sign-on through providers such as Google or Microsoft and MFA are excluded; a future production deployment should assess MFA and stronger identity controls.

3. Risks

This project combines voluntary household consumption and activity data with optional community reporting, manual data entry, transparent rule-based recommendations and gamification. For SCSA, the highest-impact risks are loss of privacy or trust, insecure handling of data, misleading results and exclusion of users. The main delivery risks are sustained engagement, feature scope and continuity after handover.

The controls below are informed by the OAIC privacy impact assessment guidance, ASD/ACSC secure software development guidance, W3C WCAG 2.2 and the Australian Government Digital Service Standard. Legal applicability should be confirmed during discovery; the guidance is used here as a prudent design baseline. Probability values are initial planning estimates (0 = impossible; 1 = certain) and should be reviewed at each sprint or milestone.

Risk	Probability (out of 1.0)	Impact	Mitigation Strategy
Privacy and re-identification. Household consumption or activity patterns, or small-community reports, may be used beyond user expectations or re-identified from supposedly aggregated results.	0.4	Very high - privacy complaints, rework and loss of trust; 60-100 hours.	Complete a privacy impact assessment before detailed design. Make records private by default; use separate, unticked opt-in choices for leaderboards and community reporting; collect only necessary data; use pseudonymous IDs; suppress small cohorts and outliers; and document retention, deletion and consent withdrawal.
Cybersecurity and service compromise. Weak code, dependencies, credentials or hosting controls could expose or alter household records, compromise accounts or interrupt the service.	0.4	Very high - data exposure or outage; 80-120 hours.	Apply secure-by-design threat modelling and OWASP ASVS. Use HTTPS, strong password hashing, least privilege, server-side validation, parameterised queries, secure cookies, dependency scanning, protected administrator access, logs, backups and pre-release security testing. Maintain incident and data-breach response plans, and recommend MFA for any future production deployment.
Data and calculation integrity. Manual entries, inconsistent units, missing periods, duplicate or outlier records, or outdated factors may produce misleading progress, badges, recommendations or comparisons.	0.7	High - misleading outcomes and lost credibility; 30-50 hours.	Use explicit units and date ranges, required-field and plausibility checks, duplicate and outlier warnings, and user correction. Version calculation factors and sources, label estimates and assumptions, regression-test formulas against

Risk	Probability (out of 1.0)	Impact	Mitigation Strategy
			a controlled data set, and review factors before release.
Low adoption and sustained engagement. Repeated manual logging, difficult onboarding or poorly matched recommendations and gamification may cause households to abandon the platform after initial use.	0.7	High - insufficient participation to demonstrate outcomes; 20-40 hours.	Interview and test with diverse households, then pilot through existing SCSA community groups. Keep onboarding and logging brief; provide saved recurring entries, relevant goals and optional reminders and leaderboards. Track activation, drop-off, weekly active use and 4/8-week retention, then iterate from feedback.
Accessibility and inclusive use. Forms, dashboards, charts, badges or leaderboards may be unusable on mobile devices or with assistive technology, excluding community members.	0.5	High - user exclusion and late remediation; 30-60 hours.	Make WCAG 2.2 Level AA part of the Definition of Done. Use semantic HTML, labelled inputs, keyboard access and visible focus, sufficient contrast, text equivalents for charts and responsive layouts. Test each sprint with automated tools plus keyboard, screen-reader and zoom checks.
Scope growth and schedule slippage. The combined feature set may exceed the student team's time and capability; late additions and integration work may displace testing and quality assurance.	0.7	High - 2-3 week delay or incomplete MVP; 50-80 hours.	Agree a signed-off MVP and MoSCoW priorities. Treat goal setting, logging, progress views and privacy choices as must-have; defer richer gamification, the optional leaderboard or recommendation refinements if needed. Maintain a prioritised backlog, acceptance criteria, change control, weekly burn-up reviews and schedule contingency.
Team continuity and operational ownership. Key-person absence, uneven skills or unclear ownership after the student team leaves may delay delivery and leave the organisation unable to operate or maintain the platform.	0.5	High - 1-2 week delay and unsustainable handover; 30-60 hours.	Assign product and technical owners; map skills, pair on critical components, review code and use a shared repository. Maintain architecture notes, a deployment runbook, administrator guide and maintenance plan. Automate deployment and backups, train the host contact, and schedule a formal handover and support window.

4. Budget

Expected cash expenditure is limited to optional minor costs. Student labour is shown separately as a hypothetical planning value and will not be charged to the client; future production operating costs remain outside the project scope.

Budget Item	Estimated Cost	Justification
Student Development Time (Hypothetical)	AUD 14,400	Six students × 80 hours × AUD 30 per hour. This is a notional labour value only and will not be charged to the client.
Web Hosting / Cloud Services	AUD 0	Free-tier or university-supported services will be used for the prototype; paid production hosting is outside the project scope.

Budget Item	Estimated Cost	Justification
Database Services	AUD 0	An open-source or free-tier database will support prototype development and testing.
Development Software and Tools	AUD 0	Development will use free or open-source tools such as GitHub, Visual Studio Code and suitable web frameworks.
UI/UX Design and Prototyping Tools	AUD 0	Free versions of Figma or equivalent tools are sufficient for wireframing and interface design.
Testing Resources	AUD 0	Testing will use the team's existing computers, mobile devices and browsers, so no additional hardware is expected.
Domain Name (Optional)	AUD 25	A temporary domain may be purchased only if a public demonstration URL is required; this cost can be avoided by using a free or university URL. Production hosting remains outside scope.
Contingency	AUD 25	A small allowance covers unexpected minor deployment or hosting expenses.
Total Hypothetical Project Value	AUD 14,450	This planning value combines notional student labour with the optional domain and contingency. It is not the expected client charge.
Expected Actual Cash Cost	AUD 0-50	This is the anticipated out-of-pocket expenditure only; free-tier services should keep the actual cash cost between AUD 0 and AUD 50.

5. Roles and Responsibilities

Roles are allocated by workstream to provide clear ownership and balanced coverage. They describe primary accountability rather than exclusive ownership: all students contribute to peer review, testing and handover, and provide backup across adjacent workstreams.

Participant	Primary role	Key responsibilities	Allocation rationale
Binglin Wu	Project Manager and Client Liaison	Coordinate planning, meetings, action and decision logs, scope control and consolidated client updates.	A single coordination point keeps responsibilities, approvals and follow-up traceable.
Chih Sheng Lin	Business Analysis and UX/UI Lead	Define requirements, user stories and acceptance criteria; prepare user flows, winvreframes and accessibility reviews.	Combining discovery and interaction design translates client needs directly into usable workflows.
Rouyu Xing	Front-end Development Lead	Build responsive interfaces, forms, dashboards, chart integration and front-end validation.	Concentrated presentation-layer ownership supports a consistent desktop and mobile-web experience.
Yueqi Zhai	Back-end and API Lead	Implement authentication, business logic, APIs, privacy permissions and server-side validation.	Centralised service and access-control logic supports secure and consistent behaviour.
Yiteng Zhang	Database and Data Analytics Lead	Design the schema and migrations; maintain units, calculation factors, aggregation and reporting queries.	Linking data structures with calculation rules improves integrity, auditability and reporting consistency.
Tulong Dai	Quality Assurance, Security and Deployment Lead	Maintain test cases and defect tracking; coordinate security and	Independent quality ownership makes acceptance and

Participant	Primary role	Key responsibilities	Allocation rationale
		accessibility checks, staging deployment and release readiness.	handover readiness explicit.
Project Mentor	Academic and Professional Adviser	Review progress weekly, advise on risks and scope, and support resolution of academic or delivery issues.	Independent oversight improves decision quality while delivery accountability remains with the team.
SCSA representatives	Client and Product Owner	Confirm priorities, review prototypes, approve scope changes, participate in user acceptance testing and accept agreed deliverables.	Client-owned decisions keep the MVP aligned with organisational needs.

Communications Plan

The communication plan defines the purpose, frequency and channel for each stakeholder group. Routine coordination uses the team's agreed communication channels, while decisions concerning requirements, privacy, scope, budget or acceptance are recorded in the shared project documentation.

Stakeholder / Group	**Purpose of Communication**	**Frequency**	**Communication Method**
Project Team Members	Discuss ongoing tasks, share progress, raise issues, clarify responsibilities, and coordinate daily project activities.	As required throughout the week	WeChat group chat
Project Team Members	Review work completed during the previous week, discuss any outstanding issues, and allocate tasks and priorities for the coming week.	Weekly, every Friday	Team meeting, either face-to-face or online
Project Mentor	Provide a summary of the week's progress, discuss completed and upcoming work, and receive feedback and guidance.	Weekly, during the Thursday class session	Face-to-face discussion during class
Client Organisation – Sustainable Communities SA	Confirm and clarify project requirements, present key project progress, obtain feedback on prototypes and deliverables, and discuss changes that may affect the agreed project scope.	Fortnightly and at key approval points; additional contact as required for clarification	Email and scheduled face-to-face or online meetings; decisions recorded in meeting minutes

The Project Manager maintains meeting minutes and an action register showing owners and due dates. Technical or delivery blockers that remain unresolved for more than two working days are escalated to the Project Mentor. Decisions concerning requirements, privacy, scope, budget, priorities or acceptance are referred to the appropriate SCSA representatives and recorded before any resulting change is implemented.

Project Stakeholders

The following stakeholders have been identified as relevant to the development and use of the household sustainability web application.

Name	Role	Interest in the project
Sustainable Communities SA – Client Representatives / Management	Project client and host organisation.	Interested in ensuring that the application supports the organisation's sustainability objectives and provides a practical tool for encouraging households to reduce energy use, water consumption, and household waste. They are also concerned with usability, maintainability, project scope, and the overall value of the final solution.
Sustainable Communities SA – Staff	Potential administrators or operational users of the system.	Interested in a system that is easy to use and manage, provides clear and reliable information, and supports community sustainability programs without creating unnecessary administrative workload.
Household Users / Community Participants	Primary end users of the web application.	Interested in a simple and accessible platform. Privacy, data security, ease of use, and control over whether their results are shared are also important concerns.
Project Team	Responsible for planning, designing, developing, testing, documenting, and delivering the web application.	Interested in successfully delivering a functional system that meets the project scope. The team also has an interest in gaining practical technical and project management experience.
Project Mentor	Provides academic and professional guidance, reviews project progress, and supports the team in resolving key issues and maintaining appropriate project direction.	Interested in the quality of the project process and outcomes.
University / Course Staff	Oversee the academic requirements and assessment of the project.	Interested in ensuring that the project meets academic standards and demonstrates achievement of the required learning outcomes.

6. Deliverables and Project Evaluation Criteria

No.	Deliverable	Description
1	Requirements and UX/UI Design Package	Confirmed requirements, user stories and acceptance criteria, together with user flows, wireframes and documented responsive and accessibility decisions.

No.	Deliverable	Description
2	Functional Web Application	A responsive, integrated MVP covering the agreed user journey for goals, challenges, logging, progress monitoring and privacy management.
3	User Account and Privacy Module	Registration, login and profile management with private-by-default records, opt-in anonymous aggregation and optional pseudonymous leaderboard participation.
4	Goal, Challenge and Rule-Based Recommendation Module	Functions for personal goals, recurring challenges, recorded activities and transparent recommendations generated from predefined rules.
5	Progress Dashboard	A dashboard presenting goal completion, activity history, progress trends and documented estimates through accessible charts, progress bars and summaries.
6	Achievement and Community Reporting Features	Achievement badges and privacy-conscious aggregated reporting. A pseudonymous leaderboard is an optional stretch feature and is not required for baseline acceptance.
7	Database and Source Code	The complete application source code, database schema, migrations and configuration files required to maintain or continue developing the system.
8	Testing Report	Test cases, expected and actual results, accessibility and security checks, identified defects and the actions taken to address them.
9	User Guide	A concise guide explaining account creation, goals, challenges, data entry, dashboard use and privacy choices.
10	Technical and Handover Documentation	System architecture, technologies, data model, setup and staging deployment instructions, calculation factors, known limitations and future recommendations.
11	Final Demonstration	A presentation and live demonstration of the main user journey and evidence that the platform addresses SCSA's agreed objectives.

Project Evaluation Criteria

From the client's perspective, the success of the project will be evaluated using the following criteria:

Evaluation Area	Client Evaluation Criteria	Proposed Measure of Success
Functional completeness	The platform provides the agreed core functions for goals, challenges, activity recording, progress monitoring and privacy management.	100% of agreed Must Have acceptance tests pass; any deferred Should Have or Could Have items are documented and accepted.
Correctness	User data, progress calculations, badge conditions and dashboard results are processed and displayed accurately.	All calculation and display regression tests pass against a controlled data set with documented expected results.
Usability	Users can understand and complete the main tasks without extensive instructions.	At least five representative users test the core workflows; at least 80% complete them without facilitator intervention, and all critical usability defects are resolved.
Dashboard quality	Charts, progress indicators and summaries are clear, relevant and easy to interpret.	At least 80% of test users interpret the key displays correctly, and the client confirms that the dashboard is suitable for demonstration and review.
Privacy protection	Users have meaningful control over whether their information is private, anonymous or included in community reporting.	Each visibility state is tested; records are private by default, and aggregation or leaderboard inclusion remains disabled until the user opts in.
Data integrity and security	The system protects user accounts and prevents unauthorised access to personal information.	Authentication, authorisation and negative tests pass, with no unresolved critical or high-severity privacy or security defects at handover.
Reliability	The application performs consistently during normal use and handles invalid or missing input appropriately.	No critical defects remain unresolved and at least 90% of all agreed test cases pass before handover.

Evaluation Area	Client Evaluation Criteria	Proposed Measure of Success
Performance	Main pages and dashboard information load within a reasonable time under expected demonstration conditions.	At least 95% of measured loads for core pages complete within three seconds using the agreed demonstration data set, devices and network conditions.
Accessibility and responsiveness	The interface is readable and usable across common desktop, tablet and mobile screen sizes.	Core workflows satisfy agreed WCAG 2.2 Level AA checks through automated and manual keyboard, zoom and screen-reader testing on the agreed browsers and devices.
Alignment with client objectives	The platform encourages households to convert general sustainability advice into practical and measurable daily actions.	Client user acceptance testing maps the MVP to the agreed objectives and records sign-off, outstanding limitations and any deferred items.
Maintainability	The system can be understood, maintained and extended by future developers.	A team member who did not author the setup instructions successfully configures and stages the application from the handover documentation.
Overall client satisfaction	The final system is appropriate for demonstration, stakeholder feedback and potential future development.	The client records acceptance of the agreed deliverables and acknowledges all documented limitations and follow-up recommendations.

7. Implementation Plan

The web application will be developed using an incremental and iterative approach. The project will be divided into manageable stages, with each stage producing a defined output for review. This approach will allow the project team to monitor progress regularly, identify issues at an early stage, and refine the application based on feedback from the client, project mentor and potential users. The implementation process will consist of the following eight stages.

1. Requirements Analysis

During the initial stage, the project team will consolidate the information collected from the initial client meeting to confirm the project objectives, target users, functional requirements, privacy requirements and project constraints. Any unclear or incomplete requirements will be discussed further with the client.

User stories and measurable acceptance criteria will then be prepared for the main system functions. These functions will include user-defined sustainability goals, recurring challenges, manual activity and consumption data entry, progress dashboards, achievement badges, personalised recommendations, optional community leaderboards and privacy controls. The confirmed requirements and acceptance criteria will provide the basis for system design, development and testing.

2. System and User Interface Design

The project team will design the user interface, system architecture and database structure based on the confirmed project requirements. Wireframes or interface prototypes will be created for the key pages, including user registration and login, the progress dashboard, goal creation, activity recording, recurring challenges, achievement badges, recommendations, privacy settings and community leaderboards.

The system architecture will define how the frontend, backend, database and application services interact. The database design will identify the main data entities and their relationships, such as users, goals, activities, challenges, achievements and privacy preferences. The proposed interface and technical designs will be presented to the client and project mentor for review before full development begins.

3. Core Platform Development

The project team will first develop the core platform and establish the basic connection between the frontend, backend and database. The initial functions will include user registration, secure login and logout, profile management and basic account settings.

The team will then implement the user-defined goal function. Users will be able to select a sustainability category, enter a measurable target value, choose an appropriate measurement unit and specify a target completion date. Input validation and appropriate error messages will be implemented to reduce invalid or incomplete data. Completion of this stage will provide the technical foundation for the remaining application features.

4. Sustainability Tracking and Challenge Development

Manual data-entry functions will be developed to allow users to record household energy consumption, water consumption, household waste and selected sustainability-related activities. Each record will include relevant information such as the activity category, value, measurement unit and date.

Daily and weekly recurring challenges will also be implemented. The system will calculate each user's progress by comparing recorded data with the user's selected target and, where appropriate, previous results. The calculation rules and measurement units will be defined consistently to ensure that the displayed results are understandable and meaningful.

5. Dashboard, Achievement and Recommendation Development

A progress dashboard will be developed to present user results through progress bars, summary statistics and appropriate charts. The dashboard will allow users to compare their current progress with their targets and review changes over time.

Achievement badges will be awarded when users satisfy predefined and transparent conditions, such as completing a challenge, reaching a goal or recording activities for several consecutive days. The application will also provide rule-based recommendations according to the user's selected goals, recorded activities and progress. The recommendations will focus on practical sustainability actions and will avoid presenting estimated results as exact environmental outcomes.

6. Community and Privacy Feature Development

The project team will develop an optional community leaderboard and aggregated community summary function. Users will be able to choose whether to keep their results private, contribute anonymised data to aggregated community reports, or display a selected username on the community leaderboard.

Privacy will be enabled by default for newly created accounts. Personal household information and directly identifiable information will not be displayed publicly without the user's explicit consent. Users will also be able to review and change their privacy preferences. Only the minimum data required for the selected community function will be processed or displayed.

7. Testing and Evaluation

The project team will conduct functional, usability, compatibility and basic security testing. Functional testing will verify whether each system function satisfies its agreed acceptance criteria. Compatibility testing will examine the application on selected web browsers and screen sizes. Basic security testing will include input validation, authentication, authorisation, session handling and protection against unauthorised access to user data.

A small group of representative participants, such as classmates or volunteer users, will be invited to complete common tasks, including creating a goal, recording consumption data, reviewing progress and changing privacy settings. The project team will collect task-completion results, observed usability issues and structured user feedback. Testing participants will use sample or non-sensitive data.

Identified defects and usability problems will be documented, prioritised according to their severity and impact, and corrected where feasible within the project scope. After corrections have been made, critical functions will be retested to confirm that the identified problems have been resolved.

8. Final Deployment and Project Handover

Following testing and refinement, a stable version of the web application will be prepared for demonstration and deployment in the agreed environment. Before deployment, the project team will verify the application

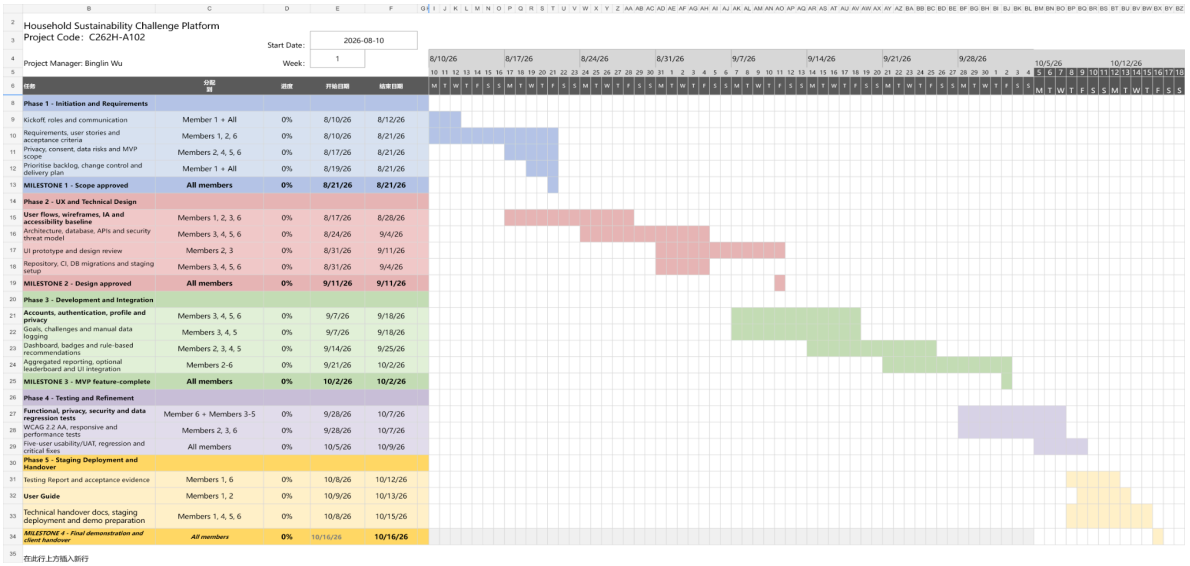
configuration, database connection and operation of critical functions. A backup or rollback approach will be prepared where technically feasible.

The final deliverables will include the application source code, database schema, system and technical documentation, user instructions, testing evidence and recommendations for future development. A final presentation and system demonstration will explain the application’s main functions and show how the completed deliverables satisfy the agreed project objectives and evaluation criteria. Any remaining limitations or incomplete features will be documented clearly during the project handover.

8. Project Schedule

The project will be delivered over a ten-week period from 10 August 2026 to 18 October 2026. The schedule is organised into five phases: initiation and requirements, UX and technical design, development and integration, testing and refinement, and staging deployment and handover. Key milestones are included to support progress monitoring and client review.

Figure 1. Project Gantt Chart and Delivery Schedule



Household Sustainability Challenge Platform
Project Code: C262H-A102
Start Date: 2026-08-10
Project Manager: Binglin Wu
Week: 1

任务	分配到	进度	开始日期	结束日期
Phase 1 - Initiation and Requirements				
Kickoff, roles and communication	Member 1 + All	0%	8/10/26	8/12/26
Requirements, user stories and acceptance criteria	Members 1, 2, 6	0%	8/10/26	8/21/26
Privacy, consent, data risks and MVP scope	Members 2, 4, 5, 6	0%	8/17/26	8/21/26
Prioritise backlog, change control and delivery plan	Member 1 + All	0%	8/19/26	8/21/26
MILESTONE 1 - Scope approved	**All members**	**0%**	**8/21/26**	**8/21/26**
Phase 2 - UX and Technical Design				
User flows, wireframes, IA and accessibility baseline	Members 1, 2, 3, 6	0%	8/17/26	8/28/26
Architecture, database, APIs and security threat model	Members 3, 4, 5, 6	0%	8/24/26	9/4/26
UI prototype and design review	Members 2, 3	0%	8/31/26	9/11/26
Repository, CI, DB migrations and staging setup	Members 3, 4, 5, 6	0%	8/31/26	9/4/26
MILESTONE 2 - Design approved	**All members**	**0%**	**9/11/26**	**9/11/26**
Phase 3 - Development and Integration				
Accounts, authentication, profile and privacy	Members 3, 4, 5, 6	0%	9/7/26	9/18/26
Goals, challenges and manual data logging	Members 3, 4, 5	0%	9/7/26	9/18/26
Dashboard, badges and rule-based recommendations	Members 2, 3, 4, 5	0%	9/14/26	9/25/26
Aggregated reporting, optional leaderboard and UI integration	Members 2-6	0%	9/21/26	10/2/26
MILESTONE 3 - MVP feature-complete	**All members**	**0%**	**10/2/26**	**10/2/26**
Phase 4 - Testing and Refinement				
Functional, privacy, security and data regression tests	Member 6 + Members 3-5	0%	9/28/26	10/7/26
WCAG 2.2 AA, responsive and performance tests	Members 2, 3, 6	0%	9/28/26	10/7/26
Five-user usability/UAT, regression and critical fixes	All members	0%	10/5/26	10/9/26
Phase 5 - Staging Deployment and Handover				
Testing Report and acceptance evidence	Members 1, 6	0%	10/8/26	10/12/26
User Guide	Members 1, 2	0%	10/9/26	10/13/26
Technical handover docs, staging deployment and demo preparation	Members 1, 4, 5, 6	0%	10/8/26	10/15/26
MILESTONE 4 - Final demonstration and client handover	***All members***	**0%**	**10/16/26**	**10/16/26**

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The editable version of the project schedule is provided separately as the Gantt chart Excel workbook.

Reference List

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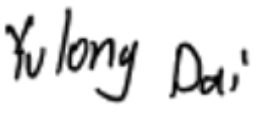
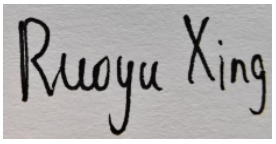
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Approval

We approve the project as described above.

Name	Role (e.g. student, client, mentor)	Signature	Date
yulong dai	student		27/08/2026
Yueqi Zhai	student		27/08/2026
yiteng zhang	student		27/08/2026
Chih Sheng Lin	student		27/08/2026
Ruoyu Xing	student		27/08/2026
Liangwei Zheng	Mentor		27/08/2026
Binglin Wu	student		27/08/2026